The existence of Smith–Toda complexes $V(4)$ for $p \geq 11$

Abstract

We construct a finite Smith–Toda complex $V(4)$ at every prime $p \geq 11$. Its Brown–Peterson homology is the canonical quotient comodule $\mathrm{BP}_*/(p, v_1, v_2, v_3, v_4)$, and it has 32 cells. The construction uses a sphere map realizing v_4 in $V(3)$. We prove that this map exists by calculating the possible Adams–Novikov differentials and applying a relation in the homotopy groups of spheres to the final target. The prime threshold is sharp by Nave's nonexistence theorem at $p = 7$.

1 Introduction

Let BP be the p-local Brown–Peterson spectrum, with Hazewinkel generators

$$\mathrm{BP}_* = \mathbb{Z}_{(p)}[v_1, v_2, \ldots], \qquad |v_i| = 2p^i - 2.$$

A Smith–Toda complex $V(n)$ realizes the quotient $\mathrm{BP}_*/(p, v_1, \ldots, v_n)$ as a finite p-local spectrum, including its canonical comodule structure and its degree-zero generator. We prove the following realization result.

Theorem 1.1. *For every prime $p \geq 11$, there is an ordinary finite p-local spectrum X_p with*

$$\mathrm{BP}_* X_p \cong \mathrm{BP}_*/(p, v_1, v_2, v_3, v_4)$$

as the canonical quotient comodule, with the isomorphism preserving the degree-zero generator. The spectrum X_p can be chosen to have 32 cells in dimensions

$$\sum_{i \in S} (2p^i - 1), \qquad S \subseteq \{0, 1, 2, 3, 4\}.$$

The prime threshold 11 is sharp. At $p = 7$, Nave's nonexistence theorem rules out the ordinary finite Smith–Toda complex $V((p+1)/2) = V(4)$ [6, Theorem 1.3]. Thus the range in Theorem 1.1 cannot include the preceding prime 7.

For $p \geq 11$, the classical complex $V(3)$ has a multiplication with a two-sided unit [4, Theorem 4.4], [3, p. 2]. It is therefore enough to realize v_4 by a sphere map into $V(3)$: multiplication then gives a v_4 self-map, whose cofiber is X_p. The preprint [1, Theorem 1.1] fixes $p = 1009$ throughout. We adapt its obstruction calculation to $p \geq 11$, proving the required uniform estimates below. The two algebraic vanishings are proved here by central extensions, and the final target is treated using a classical sphere relation.

The obstruction groups are calculated in the cobar complex using a modified May filtration and central-extension spectral sequences. After the algebraic vanishings, two differentials remain. A right-unit identity and an injective multiplication exclude the first. For the last, a divisibility calculation reduces the target to permanent classes of filtration five. A sphere relation annihilates the corresponding products, and a filtration estimate forces their vanishing before the differential can occur.

2 The Adams–Novikov realization problem

Fix a prime $p \geq 11$. All spectra and smash products below are p-local. We write

$$[i]_p = \frac{p^i - 1}{p - 1} \quad (i \geq 1), \qquad I_4 = (p, v_1, v_2, v_3) \subset \mathrm{BP}_*.$$

Thus $[i]_p = 1 + p + \cdots + p^{i-1}$ is a positive integer. The classical successive-cofiber construction gives $V(3)$ with bottom-cell map $\eta : S^0_{(p)} \to V(3)$ and canonical quotient comodule

$$\mathrm{BP}_* V(3) = \mathrm{BP}_*/I_4 = \mathbb{F}_p[v_4, v_5, \ldots]$$

[3, p. 2]. It has 16 cells in dimensions $\sum_{i \in S}(2p^i - 1)$ for $S \subseteq \{0, 1, 2, 3\}$. Choose a binary multiplication $\mu : V(3) \wedge V(3) \to V(3)$ with two-sided unit η, as supplied by [4, Theorem 4.4].

The following unit argument also closes the construction in [1, Section 7.5].

Lemma 2.1. *Suppose that $\varphi : S^{2p^4-2}_{(p)} \to V(3)$ has* BP*-Hurewicz image v_4. Then Theorem 1.1 holds.*

Proof. Define

$$f = \mu \circ (\varphi \wedge 1_{V(3)}) : \Sigma^{2p^4-2} V(3) \longrightarrow V(3), \qquad X_p = \mathrm{cofib}(f).$$

The right unit identity gives $f \circ \Sigma^{2p^4-2}\eta = \varphi$. Since $\mathrm{BP}_* V(3)$ is cyclic over BP_*, the map $\mathrm{BP}_* f$ is multiplication by v_4. This multiplication is injective, so the cofiber sequence gives

$$\mathrm{BP}_* X_p = (\mathrm{BP}_*/I_4)/(v_4).$$

The bottom-cell map induces the quotient map $\mathrm{BP}_* \to \mathrm{BP}_* X_p$; this identifies the coaction with the canonical quotient coaction and fixes the degree-zero generator. The mapping cone has the cells of $V(3)$ and their shifts by $2p^4 - 1$. These are exactly the 32 cells in the theorem. The cofiber is finite because its source and target are finite. □

Use the Adams–Novikov spectral sequence in weight grading:

$$E_2^{s,w}(V(3)) = H^{s,w} C_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_* V(3)) \Longrightarrow \pi_{2(p-1)w-s} V(3).$$

Here $C_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_* V(3))$ is the normalized cobar complex, s is cohomological degree, and $2(p-1)w$ is internal degree. The filtration-zero edge map is the BP-Hurewicz map. Thus Lemma 2.1 reduces the construction to the permanence of $v_4 \in E_2^{0,[4]_p}(V(3))$.

This spectral sequence converges to p-local homotopy [2, Theorem 2.2.13]. Its filtration is finite in each stem: the fiber of $S^0_{(p)} \to \mathrm{BP}$ has no homotopy in degrees at most zero, so the connectivity of its s-fold smash with $V(3)$ increases with s. The multiplication μ induces pairings compatible with homotopy products and the Leibniz rule; on E_2 the pairing is the cobar product [2, Theorem 2.3.3(a), (c), (e)].

Internal degrees are multiples of $2p - 2$, so a nonzero differential has degree

$$d_{2(p-1)k+1} : (s, w) \longmapsto (s + 2(p-1)k + 1, w + k), \qquad k \geq 1. \tag{2.1}$$

Its target on v_4 is a subquotient of

$$E_2^{2(p-1)k+1, [4]_p + k}(V(3)). \tag{2.2}$$

The proof treats these targets in three different ways.

For $k \geq 2$ and $k \neq p$, we prove that the target in (2.2) is already zero on E_2. The initial May calculation leaves only $k = 2, p-1$ in this range; the central-extension calculations in Section 4 eliminate both remaining positions.

For $k = 1$, the target group need not vanish. Instead, a right-unit relation puts the image of $d_{2p-1}(v_4)$ in the kernel of a multiplication map. Section 5 proves that this map is injective, so the differential itself is zero.

For $k = p$, it is enough to prove vanishing on a later page:

$$E_{2p(p-1)+1}^{2p(p-1)+1,[4]_p+p}(V(3)) = 0.$$

Section 6 divides the target classes by a large power of the beta-family detector, leaving classes of filtration five. A sphere relation makes the corresponding homotopy products zero. The bound on incoming differential lengths in Section 7 then forces these products, and hence the entire target, to vanish by the required page.

Together these conclusions exclude every outgoing differential on v_4. Filtration zero excludes incoming differentials, so permanence gives the realizing sphere map of Lemma 2.1.

We use d_r for Adams–Novikov differentials, d_r^{May} for May differentials, and ∂_r for central-extension differentials. The latter two spectral sequences calculate the cobar cohomology that forms the Adams–Novikov E_2 page.

3 Cobar comparison and the May filtration

Coordinates and cobar comparison

We use Hazewinkel generators, so that $\mathrm{BP}_*\mathrm{BP} = \mathrm{BP}_*[t_1, t_2, \ldots]$, with $|t_i| = 2p^i - 2$. The ideal I_4 is invariant, and v_4 is primitive modulo I_4 [2, Theorem 4.3.2(a),(b)]. For $i \leq 4$, the coproduct of t_i modulo I_4 is given by (3.1) [2, Corollary 4.3.15]. With coefficients BP_*/I_4, the closed degree-one cochains $[t_1^{p^j}]$ represent classes $h_{1,j}$ in cobar cohomology. Specializing p and all the v_i to zero gives the Hopf algebra

$$P = \mathbb{F}_p[t_1, t_2, \ldots], \qquad \Delta(t_i) = \sum_{\ell=0}^{i} t_\ell \otimes t_{i-\ell}^{p^\ell}, \quad t_0 = 1. \tag{3.1}$$

Write $P_r = \mathbb{F}_p[t_1, \ldots, t_r]$. We denote by $H^{s,w}(P)$ the cohomology of the normalized cobar complex $C^s(P) = \overline{P}^{\otimes s}$, where $\overline{P}$ is the positive-weight part. Write d_{cob} for the cobar differential. The specialization defines a cochain map

$$\sigma : C^\bullet_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_* V(3)) \longrightarrow C^\bullet(P)$$

that preserves bar length and weight. Its induced map on cohomology is denoted by σ_*.

The three-cochain argument below is the comparison used in [1, Lemmas 2.4 and 2.8], with the range for P_3 made explicit.

Lemma 3.1 (Cobar comparison)**.** *For $s \geq 1$, if*

$$w - s < [4]_p - 2,$$

then there are natural isomorphisms

$$E_2^{s,w}(V(3)) \cong H^{s,w}(P) \cong H^{s,w}(P_3).$$

The first is σ_*, and the second is induced by the inclusion $P_3 \hookrightarrow P$. For the target bidegrees in (2.2), the required inequality is $w - s = [4]_p - 1 - (2p-3)k < [4]_p - 2$.

Proof. Degree-s cohomology is determined by cochain degrees $d = s-1, s, s+1$. Every reduced bar factor has weight at least one. A cochain involving a positive-weight coefficient or some t_i with $i \geq 4$ has weight at least $[4]_p + d - 1 \geq [4]_p + s - 2 > w$, so no such cochain can occur. The three remaining cochain groups and their differentials involve only scalars and t_1, t_2, t_3, and hence agree. □

The carry cochain and the May spectral sequence

For $1 \leq \ell \leq p-1$, the integer $\binom{p}{\ell}$ is divisible by p. The positive carry cochain is

$$c_\beta = \sum_{\ell=1}^{p-1} \overline{\frac{1}{p}\binom{p}{\ell}} [t_1^\ell \mid t_1^{p-\ell}] \in C^{2,p}(P_1), \qquad d_{\mathrm{cob}} c_\beta = 0, \tag{3.2}$$

where the bar over the integral coefficient denotes reduction modulo p. Write $b_{1,0} = [c_\beta] \in H^{2,p}(P_1)$ and also for its image in $H^{2,p}(P)$. Its leading term will be the May generator $b_{1,0}$. Section 6 identifies this characteristic-p cochain, with the same sign, in the cobar complex for BP_*/I_4 and in the Adams–Novikov calculation.

The unweighted initial page, generator degrees, and splitting differential are the polynomial part of the classical May calculation [2, Theorem 3.2.5]. For the calculation with P_3, see also [3, authors' preprint, p. 3, equation (1)]. We use positive splitting signs, fixed in (3.11) below. The weighted family constructed next is that of [1, Lemma 3.1]. We recall its construction and convergence.

The digit filtration. The extra weight at position zero will be chosen to place possible incoming May sources below the product in the multiplication argument of Section 5. For an integer $M \geq 0$, assign the basic factor $t_i^{p^j}$ filtration

$$\lambda_{i,j} = 2i - 1 + M[j = 0], \tag{3.3}$$

where $[j = 0]$ is one for $j = 0$ and zero otherwise. If a monomial has exponents $a_i = \sum_{j \geq 0} a_{i,j} p^j$, with $0 \leq a_{i,j} < p$, put

$$\begin{aligned} F_M\Big(\prod_i t_i^{a_i}\Big) &= \sum_{i,j} a_{i,j} \lambda_{i,j}, \\ \mathcal{F}_f^M P &= \mathrm{span}_{\mathbb{F}_p}\{m : F_M(m) \leq f\}. \end{aligned} \tag{3.4}$$

Thus $F_M(t_1^{p-1}) = (p-1)(1+M)$, whereas $F_M(t_1^p) = 1$. Each carry in a product lowers filtration by

$$p\lambda_{i,j} - \lambda_{i,j+1} = (p-1)(2i-1) + pM[j=0] \geq p - 1 > 1, \tag{3.5}$$

so $F_M(mm') \leq F_M(m) + F_M(m')$. The reduced coproduct of a basic factor is

$$\overline{\Delta}(t_i^{p^j}) = \sum_{0<\ell<i} t_\ell^{p^j} \otimes t_{i-\ell}^{p^{j+\ell}}.$$

Every summand lowers filtration by exactly one, since

$$\lambda_{i,j} - \lambda_{\ell,j} - \lambda_{i-\ell,j+\ell} = 1. \tag{3.6}$$

If $j = 0$, the additional M occurs once on either side of this identity; if $j > 0$, it does not occur. Expanding the coproduct of a monomial can introduce further carries, which only lower filtration. Hence multiplication and coproduct preserve the increasing filtration $\mathcal{F}_\bullet^M P$.

The filtered cobar complex. On a cobar tensor, set

$$F_M([m_1|\cdots|m_s]) = \sum_{\nu=1}^{s} F_M(m_\nu). \tag{3.7}$$

The cobar differential splits bar factors by their reduced coproducts, so

$$d_{\mathrm{cob}}(\mathcal{F}_f^M C^{s,w}(P)) \subseteq \mathcal{F}_f^M C^{s+1,w}(P).$$

Concatenation of bar tensors also preserves filtration. The indices s and w retain their meanings as cobar degree and internal weight; f is the auxiliary May filtration degree. The resulting multiplicative spectral sequence is denoted by $\mathcal{M}_r(M)$, with page convention

$$\begin{aligned} \mathcal{M}_0^{s,w,f}(M) &= \mathcal{F}_f^M C^{s,w}(P)/\mathcal{F}_{f-1}^M C^{s,w}(P), \\ \mathcal{M}_1(M) &= H(\mathcal{M}_0(M), d_0), \\ d_r^{\mathrm{May}} &: (s,w,f) \longmapsto (s+1, w, f-r), \qquad r \geq 1. \end{aligned} \tag{3.8}$$

Here d_0 is the filtration-preserving part of d_{cob}. Write $z_{i,j}$ for the associated graded image of $t_i^{p^j}$. Equations (3.5) and (3.6) give

$$\mathrm{gr}_{\mathcal{F}^M} P \cong \bigotimes_{i\geq 1, j\geq 0} \mathbb{F}_p[z_{i,j}]/(z_{i,j}^p), \qquad \Delta z_{i,j} = z_{i,j} \otimes 1 + 1 \otimes z_{i,j}. \tag{3.9}$$

The base-p monomial basis shows that there are no further algebra relations. The page $\mathcal{M}_0(M)$ is the cobar complex of this Hopf algebra. Its differential need not vanish: for a primitive generator z, $\overline{\Delta}(z^2) = 2z \otimes z$. The exterior and polynomial generators therefore arise on its cohomology page $\mathcal{M}_1(M)$.

Lemma 3.2. *For every integer $M \geq 0$, the filtered cobar complex above gives a multiplicative May spectral sequence converging to $H^{s,w}(P)$, with finite filtration at each fixed weight. Its initial cohomology page is*

$$\mathcal{M}_1(M) = \Lambda(h_{i,j} : i \geq 1, j \geq 0) \otimes \mathbb{F}_p[b_{i,j} : i \geq 1, j \geq 0], \tag{3.10}$$

where the exterior generator $h_{i,j}$ has cohomological degree one, weight $p^j[i]_p$, and filtration $2i - 1 + M[j = 0]$. The polynomial generator $b_{i,j}$ has cohomological degree two, weight $p^{j+1}[i]_p$, and filtration $p(2i - 1 + M[j = 0])$. The symbols in this presentation are May-page generators. The leading terms of the closed cochains $[t_1^{p^j}]$ and c_β give the correspondingly named generators $h_{1,j}$ and $b_{1,0}$. The differentials have the tridegrees in (3.8), and

$$\mathcal{M}_\infty^{s,w,f}(M) \cong \mathrm{gr}_f H^{s,w}(P).$$

Put $h_{[a,b)} = h_{b-a,a}$. The first splitting differential is

$$d_1^{\mathrm{May}} h_{[a,b)} = \sum_{a<\ell<b} h_{[a,\ell)} h_{[\ell,b)}, \qquad d_1^{\mathrm{May}} b_{i,j} = 0. \tag{3.11}$$

For P_r, the same construction gives $\mathcal{M}_1(M; P_r)$ by retaining only the generators with $i \leq r$.

Proof. For a single factor $\mathbb{F}_p[z]/(z^p)$ in (3.9), factorial rescaling identifies its dual algebra with $\mathbb{F}_p[\zeta]/(\zeta^p)$. The trivial module has the periodic free resolution

$$\cdots \xrightarrow{\zeta} \mathbb{F}_p[\zeta]/(\zeta^p) \xrightarrow{\zeta^{p-1}} \mathbb{F}_p[\zeta]/(\zeta^p) \xrightarrow{\zeta} \mathbb{F}_p[\zeta]/(\zeta^p) \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

Applying $\mathrm{Hom}(-, \mathbb{F}_p)$ makes all differentials zero. The two-period shift represents multiplication by a degree-two class b, whose powers are nonzero; the degree-one class h satisfies $h^2 = 0$ by graded commutativity. Thus the cohomology ring is $\Lambda(h) \otimes \mathbb{F}_p[b]$. The class h has the weight and filtration of z. A carry representative for b is a linear combination of $[z^a|z^{p-a}]$, so both its weight and its filtration are p times those of z. For $z = z_{1,0}$, this representative is the associated graded image of c_β in (3.2). The Künneth formula gives (3.10) and the stated generator degrees.

The coproduct terms in (3.6) induce the exterior part of (3.11). For $b_{i,j}$, fix $z = z_{i,j}$. Its carry representative has total z-multiplicity p. A term lowering filtration by one splits a single z, leaving $p-1$ copies of z and two basic factors different from z. The differential d_0 preserves the multiplicity of every $z_{i,j}$. In the cohomology of the z-factor, the multiplicities of $b^a h^\epsilon$ are $pa + \epsilon$, where $a \geq 0$ and $\epsilon \in \{0, 1\}$. Since $p-1$ is not of this form, the Künneth formula makes each resulting multiplicity summand have zero d_0-cohomology. Carries lower filtration by at least $p - 1 > 1$ and cannot contribute to d_1^{May}. Hence $d_1^{\mathrm{May}} b_{i,j} = 0$.

At fixed weight w, only finitely many basic factors satisfy $p^j[i]_p \leq w$, their exponents are bounded, and bar length is at most w. The fixed-weight cobar complex is therefore finite dimensional, with bounded filtration. This proves convergence, including compatibility with multiplication. □

The abstract initial page, its (s, w) grading, and d_1^{May} are independent of M, whereas the auxiliary filtration degrees depend on M. The pages of later differentials can therefore depend on M. For the multiplication map in Section 5, equation (5.2) shows that $M = 2p$ suffices to exclude the possible incoming sources. From now on, fix

$$M = 2p, \qquad F = F_{2p}, \qquad \mathcal{M}_r = \mathcal{M}_r(2p),$$

and write $\mathcal{M}_1(P_r) = \mathcal{M}_1(2p; P_r)$. In the range of Lemma 3.1, their abutment is identified with the Adams–Novikov E_2 groups.

The low-position generators used below are

$$\begin{aligned} &\text{Degree one: } h_{1,0},\ h_{2,0},\ h_{3,0},\ h_{1,1},\ h_{2,1},\ h_{1,2};\\ &\text{Degree two: } b_{1,0},\ b_{2,0},\ b_{1,1}. \end{aligned} \tag{3.12}$$

By the general weight formulas, $h_{1,0}$ has weight 1, $h_{2,0}$ has weight $p+1$, and $h_{3,0}$ has weight p^2+p+1. The generators $h_{1,1}$ and $b_{1,0}$ both have weight p; $h_{2,1}$ and $b_{2,0}$ both have weight $p^2 + p$; and $h_{1,2}$ and $b_{1,1}$ both have weight p^2. We order the exterior generators as $h_{1,0}, h_{2,0}, h_{3,0}, h_{1,1}, h_{2,1}, h_{1,2}$. For these generators, the nonzero splitting formulas are

$$\begin{aligned} d_1^{\mathrm{May}} h_{2,0} &= h_{1,0} h_{1,1},\\ d_1^{\mathrm{May}} h_{2,1} &= h_{1,1} h_{1,2},\\ d_1^{\mathrm{May}} h_{3,0} &= h_{1,0} h_{2,1} + h_{2,0} h_{1,2}. \end{aligned} \tag{3.13}$$

Degree, weight, and column counts. To remove the contribution of powers of $b_{1,0}$, define

$$\delta(s, w) = 2w - ps. \tag{3.14}$$

Multiplication by $b_{1,0}$ does not change δ. Among all May generators, only $h_{1,0}$ has a negative contribution, namely $2-p$. The contribution of $b_{1,0}$ is zero, and every other polynomial generator contributes at least $2p(p-1)$. Consequently every initial candidate satisfies $\delta \geq 2 - p$.

The interval for $h_{i,j}$ covers the base-p positions $j, \ldots, j+i-1$; that for $b_{i,j}$ covers $j+1, \ldots, j+i$. Positions are numbered from zero. For a May monomial, let $\epsilon_{i,j} \in \{0, 1\}$ be the exponent of $h_{i,j}$

and let $m_{i,j} \in \mathbb{Z}_{\geq 0}$ be the exponent of $b_{i,j}$. Its exterior length and its exterior and total column counts are

$$\begin{aligned} m &= \sum_{i,j} \epsilon_{i,j}, \\ H_r &= \sum_{j \leq r < j+i} \epsilon_{i,j}, \\ L_r &= H_r + \sum_{j+1 \leq r < j+i+1} m_{i,j}. \end{aligned}$$

These are column counts before carrying, and they satisfy

$$s = m + 2\sum_{i,j} m_{i,j}, \qquad w = \sum_{r \geq 0} L_r p^r. \tag{3.15}$$

If $w = \sum_{r \geq 0} a_r p^r$, with $0 \leq a_r < p$, and c_r is the carry entering column r, then

$$L_r + c_r = a_r + p c_{r+1}, \qquad c_0 = 0. \tag{3.16}$$

The carries vanish for sufficiently large r. For example, the monomial

$$h_{2,0} h_{3,0} h_{1,1} h_{2,1} h_{1,2} b_{2,0}^{p-3}$$

has exterior column counts

$$(H_0, H_1, H_2, H_3) = (2, 4, 3, 0).$$

Its polynomial factor contributes $(0, p-3, p-3, 0)$ to these four columns, so

$$(L_0, L_1, L_2, L_3) = (2, p+1, p, 0).$$

Carrying from column one to column two, and then from column two to column three, gives

$$(2, p+1, p, 0) \longmapsto (2, 1, p+1, 0) \longmapsto (2, 1, 1, 1).$$

Thus $c_1 = 0$, $c_2 = c_3 = 1$, and $c_4 = 0$. The final tuple gives the base-p digits of the weight $[4]_p + 1$; the cohomological degree is $5 + 2(p-3) = 2p - 1$.

For the generators in (3.12), one has $L_2 = H_2 + m_{2,0} + m_{1,1}$, and adding their contributions gives

$$\delta = 2H_0 + p(2H_1 - m - 2m_{1,1}) + 2p^2 L_2. \tag{3.17}$$

The total degree then determines the exponent of $b_{1,0}$. Among the exterior generators in (3.12), only $h_{1,0}$ and $h_{1,2}$ fail to cover position one. For monomials in the generators of (3.12),

$$m = H_1 + \epsilon_{1,0} + \epsilon_{1,2}, \qquad -2 \leq 2H_1 - m \leq 4. \tag{3.18}$$

For odd exterior length, (3.18) gives $-1 \leq 2H_1 - m \leq 3$. For even exterior length, $H_1 - m/2 \leq 2$, with equality precisely for the exterior factor $h_{2,0} h_{3,0} h_{1,1} h_{2,1}$.

4 Vanishing in filtration $2(p-1)k + 1$

We prove

$$E_2^{2(p-1)k+1,[4]_p+k}(V(3)) = 0 \qquad (k \geq 2,\ k \neq p). \tag{4.1}$$

By Lemma 3.1, these groups are $H^{2(p-1)k+1,[4]_p+k}(P_3)$. Lemma 4.1 shows that the corresponding initial May components are zero unless $k = 2, p-1, p$. The two central-extension calculations below eliminate $k = 2, p-1$; Section 6 treats $k = p$.

For $p = 1009$, the two vanishings are [1, Corollary 6.7], proved there using the Frobenius extension. The central-extension proof below applies to $p \geq 11$. The following candidates are those of [1, Proposition 5.5, Table 3]; their enumeration below is uniform for $p \geq 11$.

Lemma 4.1. *For each integer $k \geq 2$, consider the May-page component*

$$\mathcal{M}_1^{2(p-1)k+1,[4]_p+k}(P_3),$$

where the auxiliary May filtration degree is summed over. Its nonzero cases have the following $\mathbb{F}_p$-bases.

For $k = 2$, the basis consists of

$$h_{1,0}h_{2,0}h_{3,0}b_{2,0}b_{1,0}^{p-2}b_{1,1}^{p-2},$$
$$h_{1,0}h_{2,0}h_{3,0}h_{1,1}h_{1,2}b_{1,0}^{p-2}b_{1,1}^{p-2}.$$

For $k = p-1$, the basis consists of

$$h_{1,1}b_{2,0}^3b_{1,0}^{(p-1)^2-3},$$
$$h_{2,1}b_{2,0}^2b_{1,0}^{(p-1)^2-2},$$
$$h_{1,1}h_{2,1}h_{1,2}b_{2,0}b_{1,0}^{(p-1)^2-2}.$$

For $k = p$, the basis consists of

$$h_{3,0}h_{1,1}h_{2,1}b_{1,0}^{p(p-1)-1},$$
$$h_{2,0}h_{1,1}h_{2,1}b_{2,0}b_{1,0}^{p(p-1)-2}.$$

For every other $k \geq 2$, the component is zero.

Proof. For these degrees and weights, $\delta = 2[4]_p - p - 2(p^2-p-1)k$. The inequality $\delta \geq 2-p$ gives $(p^2-p-1)k \leq [4]_p - 1$, hence $k \leq p+2$, since

$$(p^2-p-1)(p+3) - ([4]_p - 1) = p^2 - 5p - 3 > 0.$$

If any generator reaches position three, its δ contribution, even after adding $h_{1,0}$, is at least $2p^3 - 3p + 2$. For $k \geq 2$, however, the target satisfies $\delta \leq 2p^3 - 2p^2 + 5p + 6$, which is smaller. Thus only the generators in (3.12) occur.

The exterior length m is odd, so $m = 1, 3, 5$. Reduction modulo p gives $H_0 \equiv 1 + k$, with $0 \leq H_0 \leq 3$. We therefore need only consider $k = 2$ or $k = p + d - 1$ with $d = 0, 1, 2, 3$.

If $k = 2$, then $H_0 = 3$, and the exterior factor is $h_{1,0}h_{2,0}h_{3,0}$ times one of

$$1, \qquad h_{1,1}h_{2,1}, \qquad h_{1,1}h_{1,2}, \qquad h_{2,1}h_{1,2}.$$

Thus $2H_1 - m$ is either one or three. Substituting δ into (3.17) gives

$$2p(L_2 - p + 1) = 2m_{1,1} + 5 - (2H_1 - m).$$

Using $m_{1,1} \leq L_2$, we obtain $1 \leq p(L_2 - p + 1) \leq L_2 + 2$. The integer L_2 therefore satisfies

$$p \leq L_2 \leq p + \frac{2}{p-1} < p + 1.$$

Hence $L_2 = p$. Substituting the four exterior factors and using $m_{2,0} = L_2 - H_2 - m_{1,1}$, we find that adjoining either $h_{1,1}h_{2,1}$ or $h_{2,1}h_{1,2}$ gives $m_{2,0} = -1$. With no additional generator or with the additional factor $h_{1,1}h_{1,2}$, the pairs $(m_{2,0}, m_{1,1})$ are $(1, p-2)$ and $(0, p-2)$, respectively. The exponent of $b_{1,0}$ is $p-2$ in both cases, giving the two $k = 2$ entries.

If $k = p + d - 1$, then $H_0 = d$, and the counting identity becomes

$$2p(L_2 - 3 + d) = 2d + 1 + 2m_{1,1} - (2H_1 - m).$$

Using $0 \leq m_{1,1} \leq L_2$ and the odd-length case of (3.18), we get $-1 \leq p(L_2 - 3 + d) \leq L_2 + d + 1$. Thus the integer L_2 satisfies

$$3 - d \leq L_2 \leq 3 - d + \frac{4}{p-1} < 4 - d.$$

Consequently

$$L_2 = 3 - d, \qquad 2H_1 - m = 2d + 1 + 2m_{1,1} \leq 3. \tag{4.2}$$

It follows that $d + m_{1,1} \leq 1$, so only $d = 0, 1$ remain. If $d = 1$, then $m_{1,1} = 0$ and $H_1 - \epsilon_{1,0} - \epsilon_{1,2} = 3$. Since $H_0 = 1$ implies $H_1 \leq 3$, we have $m = H_1 = 3$, and neither $h_{1,0}$ nor $h_{1,2}$ occurs. The exterior factor is therefore $h_{2,0}h_{1,1}h_{2,1}$ or $h_{3,0}h_{1,1}h_{2,1}$; $L_2 = 2$ gives $m_{2,0} = 1, 0$, respectively. If $d = 0$, only $h_{1,1}, h_{2,1}, h_{1,2}$ occur. In odd exterior length, $2H_1 - m \leq 1$, so $m_{1,1} = 0$ and $2H_1 - m = 1$. The exterior factor is $h_{1,1}$, $h_{2,1}$, or $h_{1,1}h_{2,1}h_{1,2}$; $L_2 = 3$ gives $m_{2,0} = 3, 2, 1$, respectively. Finally, the total degree determines the displayed exponents of $b_{1,0}$. □

Central extensions of P_r

The cross terms in the coproduct of the last coordinate t_r of P_r involve only lower coordinates. For $r \geq 2$, the quotient by $(t_1, \ldots, t_{r-1})$ gives the Hopf extension

$$P_{r-1} \hookrightarrow P_r \twoheadrightarrow P_r/(t_1, \ldots, t_{r-1}) = \mathbb{F}_p[t_r],$$

whose quotient has primitive coordinate t_r. Its Cartan–Eilenberg spectral sequence is

$$\mathcal{E}_2^{u,v} = H^u(P_{r-1}) \otimes H^v(\mathbb{F}_p[t_r]) \Longrightarrow H^{u+v}(P_r). \tag{4.3}$$

Here the single-variable Hopf algebra $\mathbb{F}_p[t_r]$ treats t_r as primitive. The indices v and u are the fiber and base degrees, respectively; the weight grading is suppressed. The differential ∂_j increases total degree by one, decreases fiber degree by $j - 1$, and preserves weight. The filtration is finite at fixed weight.

For $P_2 \hookrightarrow P_3 \twoheadrightarrow \mathbb{F}_p[t_3]$, we first bound the support of the $\mathcal{E}_2$ diagonal $u + v = s$. A nonzero base group $H^{u,w'}(P_2)$ requires a nonzero component of $\mathcal{M}_1(P_2)$ in the same degree and weight. Tensoring this May component with the fiber cohomology gives the generators in Lemma 4.1, so its basis lists bound the support of the entire diagonal. For $k = 2$, both monomials contain exactly one $h_{3,0}$, so the target diagonal can only be

$$h_{3,0} \cdot H^{4p-4,p^3+2}(P_2).$$

For $k = p - 1$, no candidate contains a central fiber generator, so the target diagonal can only be

$$H^{2(p-1)^2+1,[4]_p+p-1}(P_2).$$

Thus it suffices to prove that these two P_2 cohomology groups vanish.

We compute these groups using $P_1 \hookrightarrow P_2 \twoheadrightarrow \mathbb{F}_p[t_2]$. Its $\mathcal{E}_2$ page is the tensor product of the base and fiber cohomology in (4.3). We name its generators as on $\mathcal{M}_1(P_2)$: $h_{2,0}, h_{2,1}$ are fiber classes of degree one and $b_{2,0}$ is a fiber class of degree two; $h_{1,0}, h_{1,1}, h_{1,2}, b_{1,0}, b_{1,1}$ lie in the base. We use

$$\begin{aligned} \partial_2 h_{2,0} &= h_{1,0}h_{1,1}, \\ \partial_2 h_{2,1} &= h_{1,1}h_{1,2}, \\ \partial_3 b_{2,0} &= b_{1,0}h_{1,2} - b_{1,1}h_{1,1}. \end{aligned} \tag{4.4}$$

The first two formulas follow directly from the coproduct. The third is the carry transgression computed in [5, §2, p. 3024, formulas (2.3) and (2.4)]. On the entire initial page, ∂_2 preserves polynomial factors and only splits $h_{2,j}$ into $h_{1,j}h_{1,j+1}$.

Lemma 4.2 (The cubic Heisenberg differential)**.** *In the spectral sequence for $P_1 \hookrightarrow P_2 \twoheadrightarrow \mathbb{F}_p[t_2]$, there is a unit $\lambda \in \mathbb{F}_p^\times$ such that*

$$\partial_4(h_{1,1}b_{2,0}^3) = \lambda b_{2,0}b_{1,0}(b_{1,0}h_{2,1}h_{1,2} + b_{1,1}h_{1,1}h_{2,1}).$$

Proof. Let $\mathcal{U} = \mathbb{F}_p[t_\alpha, t_\beta, t_\gamma]$, with t_α, t_β primitive and $\overline{\Delta}t_\gamma = t_\alpha \otimes t_\beta$. Apply the Heisenberg differential recalled in [5, §5, p. 3032, between (5.2) and (5.3)] along the Hopf map

$$\mathcal{U} \longrightarrow P_2, \qquad t_\alpha \longmapsto t_1, \quad t_\beta \longmapsto t_1^p, \quad t_\gamma \longmapsto t_2.$$

The coordinate change, index specialization, compatible units, and descent to $\mathbb{F}_p$ are given in Appendix A. □

Both remaining calculations use the spectral sequence for $P_1 \hookrightarrow P_2 \twoheadrightarrow \mathbb{F}_p[t_2]$. Since every differential raises total degree by one and preserves weight, an earlier-page boundary representing a differential value must have its source in the bidegree of the original class. The candidates are the two $k = 2$ entries of Lemma 4.1 with $h_{3,0}$ removed, and the three $k = p - 1$ entries. Their fiber degrees will exclude the possible boundaries.

The case $k = 2$. We must show that $H^{4p-4,p^3+2}(P_2) = 0$. The corresponding component of $\mathcal{E}_2$ has the two candidates from the $k = 2$ case with $h_{3,0}$ removed. Denote these monomials by

$$M_2 = h_{1,0}h_{2,0}b_{2,0}b_{1,0}^{p-2}b_{1,1}^{p-2},$$
$$M_2' = h_{1,0}h_{2,0}h_{1,1}h_{1,2}b_{1,0}^{p-2}b_{1,1}^{p-2}.$$

Here

$$M_2' = \partial_2(h_{1,0}h_{2,0}h_{2,1}b_{1,0}^{p-2}b_{1,1}^{p-2}),$$

so only M_2 remains. The fiber degree of $h_{1,0}h_{2,0}$ is one, hence $\partial_3(h_{1,0}h_{2,0}) = 0$. By (4.4),

$$\partial_3 M_2 = h_{1,0}h_{2,0}b_{1,0}^{p-2}b_{1,1}^{p-2}(b_{1,0}h_{1,2} - b_{1,1}h_{1,1}) \neq 0.$$

The right-hand side is a ∂_2-cycle of fiber degree one with two distinct terms. If it were a ∂_2-boundary, its source would be one of the two $k = 2$ entries and would have fiber degree two. But M_2, M_2' have fiber degrees $3, 1$, respectively, so no such source exists. Thus $\partial_3 M_2$ is nonzero on $\mathcal{E}_3$ and kills M_2. This P_2 group, and hence the target diagonal in the outer central extension, is zero. Lemma 3.1 gives $E_2^{4p-3,[4]_p+2}(V(3)) = 0$.

The case $k = p-1$. We must show that $H^{2(p-1)^2+1,[4]_p+p-1}(P_2) = 0$. Its weight is zero modulo p, so $h_{1,0}, h_{2,0}$ cannot occur. The component of the inner $\mathcal{E}_2$ page has three candidates. Fiber degree counts each $h_{2,j}$ once and each $b_{2,j}$ twice; base generators contribute zero. The first candidate has fiber degree six:

$$M_{p-1} = h_{1,1}b_{2,0}^3 b_{1,0}^{(p-1)^2-3}.$$

The second has fiber degree five:

$$h_{2,1}b_{2,0}^2 b_{1,0}^{(p-1)^2-2}.$$

The third has fiber degree three:

$$M_{p-1}' = h_{1,1}h_{2,1}h_{1,2}b_{2,0}b_{1,0}^{(p-1)^2-2}.$$

Indeed, on the low-position exterior generators, $\partial_2 h_{2,1} = h_{1,1}h_{1,2}$ has cohomology basis

$$1, \quad h_{1,1}, \quad h_{1,2}, \quad h_{1,1}h_{2,1}, \quad h_{2,1}h_{1,2}, \quad h_{1,1}h_{2,1}h_{1,2}.$$

The fiber-degree-five candidate therefore has nonzero ∂_2, leaving M_{p-1}, M'_{p-1}. The fiber degree of each of $h_{1,1}h_{2,1}, h_{2,1}h_{1,2}, h_{1,1}h_{2,1}h_{1,2}$ is one, so their ∂_3 differentials vanish. Consequently

$$\partial_3\big(h_{1,1}h_{2,1}b_{2,0}^2 b_{1,0}^{(p-1)^2-3}\big) = 2M'_{p-1},$$

which kills M'_{p-1}. The ∂_3 of M_{p-1} contains $h_{1,1}h_{1,2}$, which is already a ∂_2-boundary.

By Lemma 4.2, multiplication by $b_{1,0}^{(p-1)^2-3}$ gives

$$\begin{aligned} \partial_4 M_{p-1} &= \lambda D_{p-1}, \\ D_{p-1} &= b_{1,0}^{(p-1)^2-2} b_{2,0}\big(b_{1,0}h_{2,1}h_{1,2} + b_{1,1}h_{1,1}h_{2,1}\big), \qquad \lambda \in \mathbb{F}_p^\times. \end{aligned} \tag{4.5}$$

The monomial sum D_{p-1} is nonzero on $\mathcal{E}_4$: its two terms are distinct ∂_2-cycles of fiber degree three. The three $k = p-1$ entries have fiber degrees six, five, and three. Thus D_{p-1} has no incoming ∂_2 source, which would require fiber degree four. The only possible incoming ∂_3 source, which would require fiber degree five, has already been killed by ∂_2. Moreover,

$$\big(b_{1,0}h_{1,2} - b_{1,1}h_{1,1}\big)\big(b_{1,0}h_{2,1}h_{1,2} + b_{1,1}h_{1,1}h_{2,1}\big) = 0,$$

so $\partial_3 D_{p-1} = 0$. Hence (4.5) kills M_{p-1} on $\mathcal{E}_4$. The target is zero on $\mathcal{E}_5$, so this P_2 cohomology group vanishes. The outer central extension and Lemma 3.1 give $E_2^{2(p-1)^2+1,[4]_p+p-1}(V(3)) = 0$. Together with Lemma 4.1, this proves (4.1).

5 The first target and multiplication by $h_{1,4}$

The right unit satisfies

$$\eta_R(v_5) - v_5 = v_4 t_1^{p^4} - v_4^p t_1 \pmod{I_4}$$

[2, Corollary 4.3.21]. It follows that

$$v_4 h_{1,4} - v_4^p h_{1,0} = 0 \tag{5.1}$$

in cobar cohomology with coefficients BP_*/I_4.

Lemma 3.1 identifies the target of $d_{2p-1}(v_4)$ with $H^{2p-1,[4]_p+1}(P)$. The right-unit identity (5.1) will force the differential image into the kernel of multiplication by $h_{1,4}$. We therefore prove that this kernel is zero.

Proposition 5.1. *Multiplication by $h_{1,4}$ is injective:*

$$H^{2p-1,[4]_p+1}(P) \xrightarrow{h_{1,4}} H^{2p,[5]_p+1}(P).$$

The proof adapts the carry calculation and filtration separation in [1, Lemma 5.2 and Proposition 5.3]. There $M = 3p$; here (5.2) shows that $M = 2p$ suffices for $p \geq 11$.

Multiplication by $h_{1,4}$ increases cohomological degree from $2p-1$ to $2p$ and adds weight p^4. Since $[4]_p + p^4 = [5]_p$, the product has weight $[5]_p + 1$. An incoming May differential on this product preserves that weight and raises cochain degree by one, so its source must have bidegree

$$(s, w) = (2p-1, [5]_p + 1).$$

These possible differential sources are May-page classes. They are represented in the initial component $\mathcal{M}_1^{2p-1,[5]_p+1}$, with the auxiliary filtration index suppressed; possible sources on later pages lie in its subquotients.

The two carry profiles

We use the exterior order

$$h_{1,0},\ h_{2,0},\ h_{3,0},\ h_{1,1},\ h_{2,1},\ h_{1,2}.$$

The case $n = 3$ of the next lemma describes the source of Proposition 5.1. Since $h_{1,4}$ has weight p^4 and $[4]_p + p^4 = [5]_p$, the case $n = 4$ describes possible incoming May sources at the product weight.

Lemma 5.2 (Coverage at the first and last carry)**.** *Let $n = 3$ or 4. Every initial May monomial in $(s, w) = (2p-1, [n+1]_p + 1)$ has exterior length m and coverage profile*

$$m = 2D - 1, \qquad (H_0, H_1, \ldots, H_n) = (2, D+1, D, \ldots, D, 0), \qquad 2 \leq D \leq n.$$

Its polynomial factor is $b_{n-1,0}^{p-D}$. Only $D = n$ can survive the splitting differential d_1^{May}.

Proof. Existence of a carry. Let d be the total polynomial degree. All intervals lie in $[0, n+1)$. The target has base-p digits $(2, 1, \ldots, 1)$, and there are at most $n + 1 < p$ exterior intervals starting at zero. Hence exactly two occur, and there is no carry from position zero. The degree equation is

$$m + 2d = 2p - 1.$$

Write c_i for the carry entering position i. With the total coverage L_i defined in Section 3, the digit equations are

$$L_i + c_i = 1 + pc_{i+1} \quad (1 \leq i \leq n), \qquad c_1 = c_{n+1} = 0.$$

Since $L_i \leq m + d \leq 2p - 1$, an incoming carry of at most one implies $1 + pc_{i+1} \leq 2p$, and hence $c_{i+1} \leq 1$. Thus every carry is zero or one. If all carries were zero, each interval would cover at least one position, whereas the digit equations would give

$$2p - 1 = m + 2d \leq \sum_i H_i + 2\sum_i (L_i - H_i) \leq 2 + 2n,$$

a contradiction.

The endpoint count. Let ℓ, r be the first and last positions producing a carry. Then

$$L_\ell = p + 1, \qquad L_r \geq p, \qquad H_{r+1} = L_{r+1} = 0.$$

At most two exterior intervals cover both ℓ and r. They must end at $r + 1$. For $\ell = 1$, their starting points are zero or one. For $\ell > 1$, the bound $H_{\ell-1} \leq 1$ allows at most one interval starting to the left of ℓ, together with the unique possible interval $[\ell, r+1)$. We count coverage at the first and last carry positions simultaneously. Each polynomial interval contributes at most twice. Each exterior interval contributes at most once, except for the at most two intervals just identified. The degree equation makes this upper bound equal the lower bound supplied by the carry equations:

$$2p + 1 \leq L_\ell + L_r \leq m + 2 + 2d = 2p + 1.$$

Consequences of equality. Equality forces every polynomial interval to cover both endpoints and every exterior interval to cover at least one. It also gives $c_r = 1$, so $\ell < r$. If $\ell > 1$, both zero-starting exterior intervals would cover $\ell - 1$, contrary to $H_{\ell-1} \leq 1$. Thus $\ell = 1$.

Every polynomial interval is now $[1, r+1)$, and every exterior interval ends at or before $r + 1$. The last carry leaves a digit one at $r + 1$, with no nonzero digit above it, so $r + 1 = n$. At $r = n - 1$, we have $c_r = c_{r+1} = 1$, whence $H_r = p - d$. Set $D = p - d$. The D exterior intervals covering r all end at n, so $D \leq n$. Also $m = 2D - 1 \geq 3$, so $D \geq 2$. It follows that $d = p - D \geq p - n > 1$. Since every polynomial interval covers all positions from one to $n - 1$,

the digit equations force a carry at each of these positions. This gives the stated profile and polynomial factor.

The splitting differential. It remains to exclude $D < n$. Use the interval notation $h_{[i,j)} = h_{j-i,i}$. Equality in the endpoint count forces $h_{[0,n)}$ and $h_{[1,n)}$ to occur. For $D = 2$, the unique exterior term is

$$h_{[0,2)}h_{[0,n)}h_{[1,n)}.$$

Splitting $h_{[0,2)}$ produces $h_{[0,1)}h_{[0,n)}h_{[1,2)}h_{[1,n)}$ with coefficient ± 1. Splitting either of the other factors retains $h_{[0,2)}$, so this term cannot cancel.

The remaining case is $n = 4, D = 3$. Here $H_1 = 4 = m - 1$. The unique interval not covering position one must be the third interval ending at four, namely $[2, 4)$ or $[3, 4)$. The other coverage counts give

$$\begin{aligned} u = h_{[0,4)}h_{[1,4)}\big(&\lambda_1 h_{[0,2)}h_{[1,2)}h_{[2,4)} \\ &+ \lambda_2 h_{[0,3)}h_{[1,2)}h_{[3,4)} \\ &+ \lambda_3 h_{[0,2)}h_{[1,3)}h_{[3,4)}\big) \end{aligned}$$

for every linear combination u of exterior terms. If $d_1^{\mathrm{May}}u = 0$, the unique terms omitting $h_{[0,4)}$, omitting $h_{[1,4)}$, and containing $h_{[0,4)}, h_{[1,4)}, h_{[2,3)}$, respectively, give

$$\lambda_1 + \lambda_2 = 0, \qquad \lambda_1 + \lambda_3 = 0, \qquad \lambda_1 - \lambda_2 - \lambda_3 = 0.$$

Thus $\lambda_2 = \lambda_3 = -\lambda_1$ and $3\lambda_1 = 0$. Since $p \geq 11$, all three coefficients vanish. The splitting differential preserves the polynomial factor, so these cases contribute no splitting cohomology. □

Proof of Proposition 5.1. For $n = 3$, Lemma 5.2 leaves only the profile with $H_1 = 4$, $H_0 = 2$, and $m = 5$. It forces the exterior factors $h_{2,0}, h_{3,0}, h_{1,1}, h_{2,1}, h_{1,2}$. Thus splitting cohomology at the source is at most one-dimensional, with possible leading term

$$h_{2,0}h_{3,0}h_{1,1}h_{2,1}h_{1,2}b_{2,0}^{p-3}.$$

For a nonzero source class, choose a closed cobar representative with this leading term. Multiply it by the closed one-bar cochain $[t_1^{p^4}]$ representing $h_{1,4}$. The product cochain is closed, and its leading exterior product is nonzero. Only an incoming May differential could remove that leading class.

Every such incoming source lies in $(2p - 1, [5]_p + 1)$. The case $n = 4$ of Lemma 5.2 gives polynomial factor $b_{3,0}^{p-D}$, with $2 \leq D \leq 4$ and a positive exponent. Since d_1^{May} preserves polynomial factors, it cannot hit a term containing $b_{2,0}^{p-3}$.

After splitting cohomology, only $D = 4$ remains, with polynomial factor $b_{3,0}^{p-4}$. The exterior May filtration, before the extra weight at position zero, is $2\sum_i H_i - m$. It is 14 for the product and 23 for the source. Both have two exterior intervals starting at zero. To explain the choice of working filtration $F = F_{2p}$, evaluate these same candidate monomials in the family F_M of Section 3. Each exterior part acquires $2M$, and these contributions cancel in the difference. Since

$$F_M(b_{3,0}) = p(5 + M), \qquad F_M(b_{2,0}) = p(3 + M),$$

the filtration of the source minus that of the product is

$$\begin{aligned} &(23 - 14) + p\big[(p - 4)(5 + M) - (p - 3)(3 + M)\big] \\ &\qquad = 2p^2 - 11p + 9 - pM. \end{aligned} \tag{5.2}$$

The source has one fewer zero-starting polynomial factor, so its filtration relative to the product decreases by pM. For $M = 0$, the difference is positive when $p \geq 11$, and the filtration direction

alone does not exclude an incoming differential. Returning to the working filtration $F = F_{2p}$ gives $9 - 11p < 0$. Later May pages are subquotients of the splitting-cohomology page, so they have no new filtration values. By Lemma 3.2, every d_r^{May} lowers filtration. The remaining source has lower filtration than the product and cannot hit it. The leading product therefore survives to the associated graded of cohomology. Finiteness of the May filtration makes the cohomology product nonzero, proving injectivity. □

Corollary 5.3. *In the ANSS for $V(3)$, $d_{2p-1}(v_4) = 0$.*

Proof. This is the first possible positive differential. As in [1, Proposition 5.4], apply the Leibniz rule to (5.1), then apply the algebra homomorphism σ_* to the resulting E_2 identity. All terms containing v_4 vanish, leaving

$$\sigma_*(d_{2p-1}(v_4))h_{1,4} = 0.$$

Proposition 5.1 gives $\sigma_*(d_{2p-1}(v_4)) = 0$. The comparison map is an isomorphism on this target by Lemma 3.1, so $d_{2p-1}(v_4) = 0$. □

6 Divisibility and permanence for the final target

Since t_1 is primitive modulo I_4, the formula (3.2) also defines a closed cochain

$$c_\beta \in C^{2,p}_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_*V(3)).$$

Its class $b_{1,0} \in E_2^{2,p}(V(3))$ satisfies $\sigma_*(b_{1,0}) = b_{1,0} \in H^{2,p}(P)$. Let $\alpha_1 \in \pi_{2p-3}S^0_{(p)}$ be the first alpha-family class. Normalize $\beta_1 \in \pi_{2p(p-1)-2}S^0_{(p)}$ so that its image in $\pi_{2p(p-1)-2}V(3)$ is detected by $b_{1,0} = [c_\beta]$; the sign convention is recorded in Remark 6.1. In particular, $b_{1,0}$ is a permanent cycle in the ANSS.

Remark 6.1 (Carry and beta-family signs)**.** The cochain c_β in (3.2) is the positive carry. Let β_{LR} denote the sphere class β_1 in Lee–Ravenel's convention. Its image in $V(3)$ is detected by the negative carry, and hence by $-b_{1,0}$ [3, p. 2]. Our choice is $\beta_1 = -\beta_{\mathrm{LR}}$; α_1 is unchanged. The exponents $(p-1)^2$ and $p(p-1)-2$ are even, so

$$\beta_1^{(p-1)^2} = \beta_{\mathrm{LR}}^{(p-1)^2}, \qquad (-b_{1,0})^{p(p-1)-2} = b_{1,0}^{p(p-1)-2}.$$

The sphere input is the relation

$$\beta_1^{(p-1)^2} = \alpha_1\varepsilon_p. \tag{6.1}$$

Here ε_p is a sphere class in degree $2p^2(p-1)(p-2)+1$. Ravenel records that $\alpha_1\varepsilon^{(p-2)}$ is a unit multiple of $\beta_1^{1+(p-2)p}$ [2, 2nd ed., p. 139, after Theorem 4.4.22; 3rd ed., p. 141, after Theorem 4.4.20]. Since $1 + (p-2)p = (p-1)^2$, absorbing the unit into ε_p gives (6.1). The exponent $(p-1)^2$ is even, so the sign convention of Remark 6.1 does not alter this relation.

The identities

$$([4]_p + p) - p(p(p-1)-2) = 2p^2 + 4p + 1,$$
$$2p(p-1) + 1 - 2(p(p-1)-2) = 5$$

give the bidegree obtained by dividing the final target $E_2^{2p(p-1)+1,[4]_p+p}(V(3))$ by $b_{1,0}^{p(p-1)-2}$. The exponent also splits as

$$p(p-1) - 2 = (p-1)^2 + (p-3).$$

We will prove that every final-target class is divisible by $b_{1,0}^{p(p-1)-2}$ and that the resulting filtration-five class is permanent, with homotopy representatives annihilated by α_1. Relation (6.1) then makes the corresponding $\beta_1^{(p-1)^2}$ product zero in homotopy. The product of $b_{1,0}^{(p-1)^2}$ with the

filtration-five class has no outgoing differential, so if nonzero it must be killed by an incoming differential. Its filtration is $2(p-1)^2+5 < 2p(p-1)+1$, which forces this to happen before the final possible differential on v_4. Multiplying by the remaining $b_{1,0}^{p-3}$ then eliminates the whole target.

Filtration five is useful because $5 < 2p-1$ excludes incoming ANSS differentials on the remaining class. Its reduced weight $2p^2+4p+1$ also simplifies the support calculation in Lemma 6.3.

The next proof follows the filtered division in [1, Lemma 5.6(2) and equation (7.2)], with comparison bounds verified for $p \geq 11$.

Lemma 6.2 (Divisibility on the E_2 page)**.** *Multiplication gives a surjection*

$$b_{1,0}^{p(p-1)-2} : E_2^{5,\,2p^2+4p+1}(V(3)) \twoheadrightarrow E_2^{2p(p-1)+1,\,[4]_p+p}(V(3)).$$

Proof. The closed carry cochain c_β of (3.2) represents $b_{1,0}$. By the two $k=p$ entries of Lemma 4.1, dividing the initial May monomials by $b_{1,0}^{p(p-1)-2}$ leaves

$$h_{3,0}h_{1,1}h_{2,1}b_{1,0}, \qquad h_{2,0}h_{1,1}h_{2,1}b_{2,0},$$

both in $(5, 2p^2+4p+1)$.

A filtered cochain injection. The closed cochain c_β gives the degree-shifted cochain map

$$L_{c_\beta^{p(p-1)-2}} : C^{\bullet-2(p(p-1)-2),\,*-p(p(p-1)-2)}(P) \longrightarrow C^{\bullet,*}(P),$$
$$\xi \longmapsto c_\beta^{p(p-1)-2}\xi.$$

Powers and products of c_β use the cobar cup product. This map concatenates a fixed nonzero tensor on the left, whose terms all have the same May filtration. After the corresponding filtration shift, it is a strict filtered injection. The quotient sequence is therefore exact on associated graded complexes.

Vanishing of the quotient. Consider the quotient complex

$$C^\bullet(P)/c_\beta^{p(p-1)-2}C^\bullet(P).$$

On the initial May page, multiplication by $b_{1,0}^{p(p-1)-2}$ is injective in every bidegree, because $b_{1,0}$ is a polynomial variable. The associated graded long exact sequence identifies the quotient's initial May page with this cokernel. The cokernel is zero at the target by the monomial calculation. The finite May filtration then gives zero quotient cohomology there.

Division by a cocycle. Thus every target cocycle ζ can be written

$$\zeta = c_\beta^{p(p-1)-2}\xi + d_{\mathrm{cob}}\rho,$$
$$\xi \in C^{5,\,2p^2+4p+1}(P), \qquad \rho \in C^{2p(p-1),\,[4]_p+p}(P).$$

Applying d_{cob} gives $c_\beta^{p(p-1)-2}d_{\mathrm{cob}}\xi = 0$. The cochain injection implies $d_{\mathrm{cob}}\xi = 0$, and hence $[\zeta] = b_{1,0}^{p(p-1)-2}[\xi]$ in $H^{2p(p-1)+1,[4]_p+p}(P)$.

The specialization cochain map preserves c_β, so σ_* commutes with multiplication by $b_{1,0}$. Since

$$(2p^2+4p+1)-5 < [4]_p - 2,$$

Lemma 3.1, applied to source and target, transfers the surjection to the ANSS E_2 page for $V(3)$. □

It remains to prove permanence and α_1-annihilation for the filtration-five source in Lemma 6.2. We use the small-stem calculation of [1, Lemma 7.1], replacing its numerical estimates by bounds valid for $p \geq 11$.

Lemma 6.3 (Permanence and multiplication by α_1)**.** *Every class in $E_2^{5,\,2p^2+4p+1}(V(3))$ is a permanent cycle. For every homotopy representative of such a class,*

$$\gamma \in \pi_{4p^3+4p^2-6p-7}V(3),$$

one has $\alpha_1\gamma = 0$.

Proof. Within this proof, set $w_0 = 2p^2 + 4p + 1$. Permanence follows if all targets of $d_{2(p-1)j+1}$ on $E_2^{5,w_0}(V(3))$ vanish. Since $|\alpha_1| = 2p-3$, the assertion about multiplication by α_1 follows from

$$\pi_{2(p-1)(w_0+1)-6}V(3) = 0. \tag{6.2}$$

An outgoing $d_{2(p-1)j+1}$ from $(5, w_0)$ ends at $(6+2(p-1)j, w_0+j)$, with $j \geq 1$. In the stem (6.2), the equation $2(p-1)w - s = 2(p-1)(w_0+1) - 6$ gives $s \equiv 6 \pmod{2p-2}$. Since $2p-2 > 6$, its nonnegative filtrations are $s = 6 + 2(p-1)j$ with $j \geq 0$, and then $w = w_0 + 1 + j$. Thus both required vanishings are covered by the families

$$(s, w) = (6 + 2(p-1)j, w_0 + \varepsilon + j), \qquad \begin{cases} \varepsilon = 0, & j \geq 1, \\ \varepsilon = 1, & j \geq 0. \end{cases} \tag{6.3}$$

They satisfy

$$w - s = w_0 + \varepsilon - 6 - (2p-3)j \leq w_0 - 5 < [4]_p - 2,$$

so Lemma 3.1 applies throughout. For $j \geq 3$, the bound $\delta \leq -2p^2 + 8p + 10 < 2 - p$ contradicts (3.14). Only $j = 0, 1, 2$ remain. Here $w < 3p^2$, so the available generators are

$$h_{1,0}, h_{2,0}, h_{3,0}, h_{1,1}, h_{2,1}, h_{1,2}, \qquad b_{1,0}, b_{2,0}, b_{1,1}.$$

Let m be the even exterior length. In this range the total coverage at position two is $L_2 = H_2 + m_{2,0} + m_{1,1}$. Reduction modulo p and (3.17) give

$$H_0 = 1 + \varepsilon + j, \qquad p(L_2 + j - 2) = 1 + j + m/2 + m_{1,1} - H_1.$$

Since $m_{1,1} \leq L_2$, (3.18) gives $-1 \leq p(L_2 + j - 2) \leq L_2 + j + 2$. The integer L_2 therefore satisfies

$$2 - j \leq L_2 \leq 2 - j + 4/(p-1) < 3 - j.$$

Consequently

$$L_2 = 2 - j, \qquad H_1 - m/2 = 1 + j + m_{1,1}. \tag{6.4}$$

The cases $j \geq 1$. By (3.18), $H_1 - m/2 \leq 2$, with equality only for $h_{2,0}h_{3,0}h_{1,1}h_{2,1}$. Equation (6.4) then forces $j = 1$ and $m_{1,1} = 0$. But this exterior term has $H_2 = 2$, contrary to $L_2 = 2 - j = 1$.

The bidegree $(6, w_0 + 1)$. For $j = 0, \varepsilon = 1$, we have $H_0 = 2$. If $m_{1,1} \geq 1$, equality would force $h_{2,0}h_{3,0}h_{1,1}h_{2,1}$ and $m_{1,1} = 1$, giving $H_2 + m_{1,1} = 3 > L_2 = 2$. Thus $m_{1,1} = 0$ and $H_1 = m/2 + 1$. Since $H_0 = 2$, the exterior length is two or four. Completing the polynomial exponents gives exactly four candidates. For those containing $b_{2,0}$, (3.13) gives

$$d_1^{\mathrm{May}}(h_{2,0}h_{3,0}b_{1,0}b_{2,0}) = (h_{1,0}h_{2,0}h_{2,1} - h_{1,0}h_{3,0}h_{1,1})b_{1,0}b_{2,0} \neq 0,$$
$$d_1^{\mathrm{May}}(h_{1,0}h_{2,0}h_{1,1}h_{2,1}b_{2,0}) = 0.$$

The two candidates with polynomial factor $b_{1,0}$ have equal nonzero differentials:

$$d_1^{\mathrm{May}}(h_{1,0}h_{3,0}h_{1,1}h_{2,1}b_{1,0}) = -h_{1,0}h_{2,0}h_{1,1}h_{2,1}h_{1,2}b_{1,0},$$
$$d_1^{\mathrm{May}}(h_{2,0}h_{3,0}h_{1,1}h_{1,2}b_{1,0}) = -h_{1,0}h_{2,0}h_{1,1}h_{2,1}h_{1,2}b_{1,0}.$$

The nonzero differential with polynomial factor $b_{1,0}b_{2,0}$ cannot cancel with either value having polynomial factor $b_{1,0}$. The cycles are therefore spanned by

$$h_{1,0}h_{2,0}h_{1,1}h_{2,1}b_{2,0},$$
$$(h_{2,0}h_{3,0}h_{1,1}h_{1,2} - h_{1,0}h_{3,0}h_{1,1}h_{2,1})b_{1,0}.$$

Both are boundaries:

$$h_{1,0}h_{2,0}h_{1,1}h_{2,1}b_{2,0} = -d_1^{\mathrm{May}}(h_{2,0}h_{3,0}h_{1,1}b_{2,0}),$$
$$(h_{2,0}h_{3,0}h_{1,1}h_{1,2} - h_{1,0}h_{3,0}h_{1,1}h_{2,1})b_{1,0} = d_1^{\mathrm{May}}(h_{2,0}h_{3,0}h_{2,1}b_{1,0}).$$

Hence splitting cohomology at this position is zero. By the finite May filtration and Lemma 3.1, $E_2^{6,w_0+1}(V(3)) = 0$.

The shortest topological differential has length $2p-1 > 5$, so none can enter filtration five. The $\varepsilon = 0$ calculation eliminates all outgoing targets, and every class in $E_2^{5,w_0}(V(3))$ is therefore permanent. The $\varepsilon = 1$ calculation makes every associated graded group of $\pi_{2(p-1)(w_0+1)-6}V(3)$ vanish. Finite convergence proves (6.2), and thus $\alpha_1\gamma = 0$ for every homotopy representative. □

7 Completion of the realization

We use the finite-page argument of [1, Proposition 7.8], with its homotopy input supplied here by (6.1). Suppose a product of permanent cycles has filtration $f \geq 1$ and the corresponding homotopy product is zero. Compatibility with homotopy makes its E_∞ class zero, while the Leibniz rule excludes outgoing differentials wherever the product is nonzero. If it is nonzero on E_2, an incoming d_r must therefore kill it. Since d_r raises filtration by r and the source has nonnegative filtration, $r \leq f$. The product is thus zero by E_{f+1}.

Proof of Theorem 1.1. Equation (4.1) excludes every differential $d_{2(p-1)k+1}(v_4)$ with $k \geq 2$, $k \neq p$, and Corollary 5.3 excludes $k = 1$. It remains to consider $d_{2p(p-1)+1}(v_4)$.

Lemma 6.2 writes every class in $E_2^{2p(p-1)+1,[4]_p+p}(V(3))$ as

$$b_{1,0}^{p(p-1)-2}\theta = b_{1,0}^{p-3}(b_{1,0}^{(p-1)^2}\theta), \qquad \theta \in E_2^{5,2p^2+4p+1}(V(3)).$$

By Lemma 6.3, the class θ is permanent, and each homotopy representative $\gamma \in \pi_{4p^3+4p^2-6p-7}V(3)$ satisfies $\alpha_1\gamma = 0$. The sphere action and (6.1) give

$$\beta_1^{(p-1)^2}\gamma = \pm\varepsilon_p(\alpha_1\gamma) = 0.$$

By the left unit identity for μ and compatibility of the ANSS pairing with homotopy, the E_∞ image of $b_{1,0}^{(p-1)^2}\theta$ is the associated graded class of $\beta_1^{(p-1)^2}\gamma$, which is zero by the homotopy relation just proved. Both $b_{1,0}$ and θ are permanent, so the finite-page observation applies. Its filtration is $2(p-1)^2+5$, and

$$2p(p-1)+1-(2(p-1)^2+5) = 2(p-3) > 0.$$

Thus the product is zero by $E_{2(p-1)^2+6}$, hence by $E_{2p(p-1)+1}$. Multiplication by $b_{1,0}^{p-3}$ shows that every class in the target of $d_{2p(p-1)+1}(v_4)$ is zero on that page.

The class v_4 is therefore permanent. Finite convergence and the filtration-zero edge map produce $\varphi : S_{(p)}^{2p^4-2} \to V(3)$ with BP-Hurewicz image v_4. Lemma 2.1 gives X_p with the asserted comodule structure and cell dimensions. □

A The cubic Heisenberg differential

Proof of Lemma 4.2. We use the central extension of the algebraic Heisenberg group in [5, §5, p. 3032, between formulas (5.2) and (5.3)]. For a lower triangular unipotent matrix, the coordinate functions satisfy $\overline{\Delta} a_{31} = a_{32} \otimes a_{21}$. Set

$$t_\alpha = a_{21}, \qquad t_\beta = a_{32}, \qquad t_\gamma = a_{21}a_{32} - a_{31}.$$

Then t_α, t_β are primitive and

$$\overline{\Delta} t_\gamma = a_{21} \otimes a_{32} = t_\alpha \otimes t_\beta.$$

On the center, $t_\gamma = -a_{31}$. Thus the fiber exterior and carry classes are multiplied by the unit -1; these compatible changes of sign and normalization are absorbed into λ.

In these coordinates, take $\mathcal{U} = \mathbb{F}_p[t_\alpha, t_\beta, t_\gamma]$ with the coproduct just specified. There is a coproduct-preserving map

$$\mathcal{U} \longrightarrow P_2, \qquad t_\alpha \longmapsto t_1, \quad t_\beta \longmapsto t_1^p, \quad t_\gamma \longmapsto t_2.$$

In Yagita's notation, the exterior and polynomial generators are related by the cobar Bockstein $y(i+1) = \widetilde{\beta} x(i)$. Take $i = 1$, $j = 1$, and $\xi = \alpha$ in its formula for $\partial_4(x_\xi(i) y_\gamma(i)^{j+2})$. The four base classes $(x_\alpha(1), x_\beta(1), y_\alpha(1), y_\beta(1))$ map to $(h_{1,1}, h_{1,2}, b_{1,0}, b_{1,1})$, and the two central classes map to unit multiples of $h_{2,1}, b_{2,0}$. Its right-hand side specializes, up to these compatible units, to

$$\begin{aligned} &(b_{1,0}h_{1,2} - b_{1,1}h_{1,1}) b_{1,0} h_{2,1} b_{2,0} \\ &\qquad = -b_{1,0} b_{2,0} (b_{1,0} h_{2,1} h_{1,2} + b_{1,1} h_{1,1} h_{2,1}). \end{aligned}$$

After choosing signs consistently with (4.4), naturality gives the asserted differential. The coefficient λ absorbs the choices of sign and normalization. The assumption $p \geq 11$ ensures that the coefficients needed for the cubic formula are invertible. Yagita works over an extended coefficient field. Cobar cohomology commutes with field extension, so the displayed differential holds over $\mathbb{F}_p$. □

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